OPENSOURCESDRLAB DOCUMENTATION

FMC-AD9371 Quick Start Guide

Product	FMC-AD9371
Document	OSSDRLAB-FMC-AD9371-QSG
Version	1.0
Date	2026-11-12

1. Overview

The FMC-AD9371 is an FPGA Mezzanine Card (FMC, HPC connector) built around the Analog Devices AD9371 RF transceiver and the AD9528 clock generator. It brings the transceiver's two transmit, two receive and two observation-receive channels out to SMA connectors and connects to the carrier FPGA through JESD204B. The wiring follows the Analog Devices AD9371 evaluation card (without its GPIO header).

This guide shows how to run the prebuilt bare-metal reference designs on a Xilinx ZC706 or on the XC7Z100 "7100" carrier, how to look at the transmit and receive data in Vivado, and how to rebuild the designs from the sources in this repository.

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2. What you need

Item	Notes
FMC-AD9371 board	
Carrier board	Xilinx ZC706 (XC7Z045), or the XC7Z100 "7100" carrier
JTAG and USB-UART cables	for programming and for the serial console
30.72 MHz reference clock (optional)	connected to REF_IN; see the tip in section 4
SMA cable and attenuator (optional)	for a TX-to-RX loop-back test
PC with Vivado and Vitis 2021.1	Vivado Lab Edition is enough to program the board and use the ILA; download from AMD: https://www.xilinx.com/support/download.html
Prebuilt files	`FMC-AD9371_<design>_prebuilt_Vivado2021.1.zip` from the Releases page
Serial terminal program	any terminal program

The repository contains three designs. All use a 122.88 MHz device clock and the same FPGA logic; they differ in the AD9371 radio profile and in the carrier.

Design	Carrier	Receive / observation receive	Transmit	RF LO set by the application
`ZC706_122M88`	ZC706	100 MHz bandwidth, 122.88 MSPS	122.88 MSPS	6 GHz
`ZC706_MAX`	ZC706	100 MHz bandwidth, 153.6 MSPS	250 MHz bandwidth, 307.2 MSPS	6 GHz
`7100_MAX`	7100	100 MHz bandwidth, 153.6 MSPS	250 MHz bandwidth, 307.2 MSPS	3 GHz

`MAX` designs run the AD9371 at its maximum data rates.

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3. Board connectors

Connector	Function
TX1, TX2	transmit outputs
RX1, RX2	receive inputs
ORX1, ORX2	observation-receiver inputs
T_LO, R_LO	external LO ports of the transmit and receive synthesizers
REF_IN	external reference clock input of the AD9528 (30.72 MHz in these designs)
STATUS LEDs	AD9528 status outputs
FMC (HPC)	JESD204B data, SPI control and 12 V power from the carrier

See `hardware/FMC-AD9371_Schematic.pdf` for the full schematic.

4. Run a prebuilt design

1. With the carrier switched off, fit the FMC-AD9371 on the carrier's **HPC** FMC connector. Connect the JTAG cable and the USB-UART cable to the PC. If you have one, connect a 30.72 MHz reference clock to **REF_IN**.
2. For a loop-back test, connect **TX1** to **RX1** through an SMA cable and an attenuator (for example 30 dB). Never connect a transmit output directly to a receive input.
3. Set the carrier's boot-mode switches to **JTAG** and power it on. Open the serial console (115200 baud, 8N1).
4. Unzip the prebuilt files for your design, open the Xilinx Software Command-Line Tool (XSCT, installed with Vitis) in that folder and run:

```
connect
targets -set -nocase -filter {name =~ "*A9*#0"}
rst -system
after 1000
fpga -file jesd204b_adc_dac_top.bit
dow fsbl.elf
con
after 3000
stop
dow FMC_AD9371_PS.elf
con
```

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The FSBL initialises the processor system; the application then configures the AD9528 and the AD9371, loads the AD9371 ARM firmware, runs the initial calibrations and starts the JESD204B links.

5. Check the console. A successful start prints:

```
AD9528_initialize successful!
FMC_AD9371 config successful!
tx_lo_freq=6000000000Hz
rx_lo_freq=6000000000Hz
```

(3000000000 Hz with the `7100_MAX` design.)

6. In the Vivado Hardware Manager, open the target, select the FPGA and load the probes file `jesd204b_adc_dac_top.ltx` (Refresh Device if the ILA windows do not appear). Two ILA cores are available: `ila_adc_inst` shows the receive data from the AD9371, and `ila_dac_inst` shows the transmit data, test tones generated by two DDS cores in the FPGA. Run a capture to see the waveforms.

> **Tip:** without a reference clock on REF_IN the console prints *WARNING: AD9528_initialize() issues. Possible cause: REF_CLK not connected.* Connect a 30.72 MHz reference to lock the clock generator.
>
> **Tip:** to change the RF frequency, edit `tx_freq` and `rx_freq` in `FMC_AD9371_PS/src/headless.c` and rebuild the application (section 5).

Booting the 7100 carrier from an SD card

Create a `BOOT.bin` from the prebuilt files: the `7100_MAX` package contains a `boot.bif`, so in that folder run `bootgen -arch zynq -image boot.bif -o BOOT.bin -w` (bootgen comes with Vivado/Vitis; Vitis > Create Boot Image does the same). For the other designs, list the FSBL, the bitstream and the application in that order. Copy `BOOT.bin` to the root of a FAT32-formatted microSD card, insert the card, set the boot-mode switches to **SD card** and power on. The console shows the same messages as in step 5.

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5. Reference designs in this repository

Folder	Contents
`vivado/<design>/FMC_AD9371_PL/`	Vivado 2021.1 project `JESD204B_TX_RX.xpr`: block design, IP cores, constraints and HDL (`src/`)
`vivado/<design>/FMC_AD9371_PS/`	Bare-metal application sources and the exported hardware platform `jesd204b_adc_dac_top.xsa`
`FMC_AD9371_PS/ad9371/`	Board set-up: AD9371 radio profile (`myk.c`), initialisation (`rf_config.c`), JESD204 core and PHY drivers, LO change
`FMC_AD9371_PS/ad9528/`	AD9528 clock-generator driver and settings
`FMC_AD9371_PS/mykonos/`	Analog Devices AD9371 (Mykonos) API 1.5.1
`FMC_AD9371_PS/drivers/`	Analog Devices no-OS platform drivers (SPI, GPIO)
`FMC_AD9371_PS/src/`	`main()` in `headless.c`, platform parameters, linker script
`FMC_AD9371_PS/AD9371_Trans_Create/`	Profile files exported from the Analog Devices transceiver evaluation software, for reference

To rebuild from source:

1. Open `FMC_AD9371_PL/JESD204B_TX_RX.xpr` in Vivado 2021.1. If Vivado reports the block-design wrapper as missing, right-click `design_1.bd` and choose **Create HDL Wrapper** (let Vivado manage it). Generate the bitstream and export the hardware (`.xsa`, including the bitstream).
2. Download the Analog Devices Mykonos ARM firmware header `Mykonos_M3.h` into `FMC_AD9371_PS/ad9371/`. It is not part of this repository; `FMC_AD9371_PS/ad9371/README.md` gives the link (ADI no-OS, branch `2018_R1`, firmware 5.1.1) and the SHA-256 value to check.
3. In Vitis 2021.1, create a platform from the `.xsa` (standalone, `ps7_cortexa9_0`) and an empty bare-metal C application, then import the folders `ad9371`, `ad9528`, `drivers`, `mykonos` and `src` from `FMC_AD9371_PS` into the application (including `src/lscript.ld`).
4. In the application's C/C++ build settings, add these include directories (ARM v7 gcc compiler > Directories):

   ```
   ad9371
   ad9528
   drivers
   mykonos
   mykonos/mykonos_debug
   src
   ```

   The linker libraries are the Vitis defaults (`-Wl,--start-group,-lxil,-lgcc,-lc,--end-group`).

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5. Build the application (Release configuration), then run it over JTAG as in section 4, or create `BOOT.bin` to boot from an SD card.

6. Support

- Product page: https://opensourcesdrlab.com/products/opensourcesdrlab-high-performance-ad9371-fmc-board-for-fpga-mezzanine-card-compatible-with-official-version
- Email: support@opensourcesdrlab.com
- Forum: https://forum.opensourcesdrlab.com
- Discord: https://discord.gg/nnxY6DkmvT

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